

PWM

Pulse width modulation

technique to control the amount of power you provide to certain analog devices

- PWM helps control analog devices using digital circuits.

↳ Duty Cycle

$$d = \frac{t_{on}}{t_{pwm}}$$



analog device sees avg voltage over a period of time

↳ Resolution

min increment step of duty cycle

no. of bits = R.

$$\text{i.e. } R = 8 \rightarrow \text{min inc. step} = \frac{1}{2^8}$$

so duty cycle is $= \frac{1}{256}$

$$\frac{0}{256}, \frac{1}{256}, \dots, \frac{255}{256}, \frac{256}{256}$$

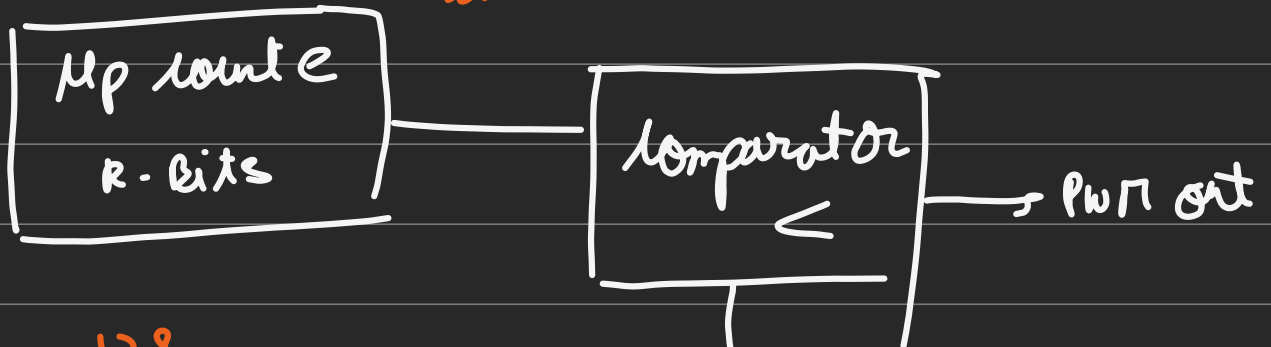
↳ Switching frequency

$$f_{\text{PWM}} = \frac{1}{t_{\text{PWM}}}$$

Basic PWM Design

$$0 \rightarrow 2^R - 1$$

$$0 \ 1 \ 2 \rightarrow 255 \text{ at } 128$$



high till 127

low at 128

→ $(V_p) < (\text{duty}) ? 1:0$

Problem: for 100% duty cycle at ass
this statement becons
full so

Something like this will occur

Enhanced PWM Design

system clock period $T_{\text{sys}} \rightarrow 100 \text{ MHz}$
(10 ns)

PWM period $T_{\text{pwm}} = 2^R \times T_{\text{sys}} \times$
(T final value + 1)

switching $f = \frac{1}{T_{\text{pwm}}}$

problem with previous "if $R \uparrow$ then no
of Reg 1, resolution changes

so

Timer



Up Counter

by solving

$$\text{Timer final value} = \frac{1}{2^8 \times (T_{sys} \times f_{PWM})}$$

i.e

$$\frac{1}{2^8 \times (10 \times 10^{-9} \times 2000)} - 1$$

$$= 194.312$$

final Design

Timer



Up Counter

Up Counter

Up Counter



cause

this is combinational

this is sequential

while transition

if it happens that while increment it touches that value which is greater than duty, might cause glitches

(R2) if comp is faster than counter then might see



